

NASSAU CANDY DISTRIBUTOR

PRODUCT LINE PROFITABILITY & MARGIN

PERFORMANCE ANALYSIS

An End-to-End Data Analytics Project

Research Paper

Prepared by

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Submitted for

Unified Mentor Pvt. Ltd. Internship Project

Project Domain: Business Intelligence & Profitability Analytics

Tools: Microsoft Excel & MySQL & Microsoft Power BI

Dataset Size: 10,194 Transaction Records

Total Sales: \$141,783.63

Total Gross Profit: \$93,442.80

Overall Profit Margin: 65.91%

DECLARATION

I, **Sonu Kumar**, hereby declare that this research paper titled “Nassau Candy Distributor – Product Line Profitability & Margin Performance Analysis” is prepared as part of my internship project with Unified Mentor Pvt. Ltd. I declare that the analytical work presented in this paper is based on the supplied Nassau Candy Distributor project dataset and the analytical workflow implemented using Microsoft Excel, MySQL, and Microsoft Power BI. The calculations, tables, interpretations, visualizations, and recommendations have been prepared for academic and professional learning purposes. I further declare that the empirical findings reported in this paper are derived from the project data and are not represented as results from external commercial datasets. External references are used only to provide conceptual and methodological context for data analytics, profitability analysis, SQL, Excel, business intelligence, and dashboard design. Where project limitations exist, they are stated explicitly. The study is descriptive and observational; therefore, the findings indicate observed patterns and relationships within the supplied data and should not be interpreted as proof of causal effects.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to Unified Mentor Pvt. Ltd. for providing me with the opportunity to undertake this project as part of my internship experience. The project gave me an opportunity to apply classroom knowledge to a practical business analytics problem and to understand how transactional data can be transformed into profitability insights. I am thankful to the mentors, coordinators, and reviewers associated with the internship for providing a structured environment in which the project could be completed. Their guidance helped me improve the analytical workflow, documentation, interpretation of results, and presentation of findings. I also acknowledge the role of Microsoft Excel, MySQL, and Microsoft Power BI in enabling the different stages of this work. Excel was used for data inspection, calculations, exploratory analysis, and supporting summaries. MySQL was used to formulate structured business questions and reproduce profitability results. Power BI was used to translate validated analytical results into an interactive dashboard. A particularly valuable part of the project was cross-validating the same business metrics across multiple tools. The workflow helped demonstrate that a dashboard should not be treated as an isolated presentation layer; its headline values should remain traceable to source data and analytical definitions. Finally, I thank my family, teachers, friends, and everyone who supported me during the completion of this work. This project has strengthened my interest in data analytics and improved my practical understanding of data preparation, exploratory analysis, SQL querying, dashboard development, business interpretation, and technical documentation.

Sonu Kumar

ABSTRACT

This research paper presents an end-to-end descriptive and diagnostic analysis of 10,194 Nassau Candy Distributor transaction records using Microsoft Excel, MySQL, and Microsoft Power BI. The study focuses on product-line profitability, regional gross-profit performance, sales, cost, units sold, and product-level profit margin. The purpose is to transform raw transactional data into an auditable analytics workflow and a decision-oriented business intelligence dashboard. The methodology begins with source-data inspection and validation in Excel. Exploratory summaries are then used to establish baseline financial measures and identify important product, division, and regional patterns. The dataset is subsequently analyzed through structured SQL queries in MySQL. Finally, Power BI measures, visuals, and slicers are used to communicate the validated results. The dataset contains 10,194 transaction records and 8,549 distinct orders covering 2 January 2024 to 31 December 2025.

Total sales are \$141,783.63, total cost is \$48,340.83, total gross profit is \$93,442.80, and total units sold are 38,654. The overall gross-profit margin is 65.91%.

Profitability is highly concentrated in the Chocolate division, which generates \$88,824.62 in gross profit. Pacific is the leading region with \$30,485.94. The top five products by gross profit are all Wonka Bar products, led by Wonka Bar - Scrumdiddlyumptious with \$19,357.50 gross profit.

The analysis recommends monitoring product concentration, protecting strong product availability, investigating regional differences, and evaluating products using both absolute gross profit and margin. Because the study is descriptive, it identifies observed patterns rather than causal effects or future forecasts. Keywords: Nassau Candy Distributor, Profitability Analysis, Gross Profit, Profit Margin, Sales Analytics, Excel, MySQL, SQL, Power BI, Business Intelligence, Data Visualization, Dashboard.

EXECUTIVE SUMMARY

The project converts a raw transactional dataset into a validated three-tool analytics solution. The workflow is Excel → MySQL → Power BI and is designed around the business questions of profitability, product contribution, regional performance, and margin Metric Value.

Total transaction records: 10,194

Unique orders: 8,549

Total sales: \$141,783.63

Total cost: \$48,340.83

Total gross profit: \$93,442.80

Total units sold: 38,654

Overall profit margin: 65.91%

The analysis produces five practical conclusions. First, profitability is strongly concentrated in the Chocolate division. Second, Pacific is the strongest region in absolute gross profit. Third, the top five products by gross profit are all Wonka Bar products. Fourth, the leading products show different margin percentages, demonstrating why sales ranking alone is insufficient. Fifth, the Power BI dashboard provides a compact interactive layer for date, region, and division investigation. The research contribution is therefore a reproducible analyst workflow rather than a new predictive algorithm. The project demonstrates how raw transaction data can move through validation, structured SQL analysis, interactive dashboard reporting, and business recommendations.

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1. INTRODUCTION

Distribution businesses operate across large product portfolios, geographic markets, customers, and transaction volumes. In such environments, revenue alone is not sufficient to evaluate performance. Management must understand how sales translate into cost and gross profit and which product or regional segments contribute most strongly to the overall result. The Nassau Candy Distributor dataset provides a suitable context for this type of analysis because it combines transaction identifiers, order dates, product information, division, region, units, sales, cost, and gross profit. These dimensions allow an analyst to move beyond a single revenue number and investigate the structure of profitability. This study treats gross profit as the central business outcome and uses sales, cost, units, division, region, and product as supporting dimensions. The analysis is descriptive and observational. It identifies observed patterns in the supplied dataset but does not claim that a particular product, region, or division causes higher or lower profitability.

1.1 BUSINESS CONTEXT AND PROFITABILITY ANALYTICS

Profitability analytics helps organizations identify where economic value is being generated and where management attention may be required. A product can have high sales but a lower margin, while another product can have a smaller revenue contribution but a stronger percentage margin. Therefore, absolute gross profit and relative margin should be interpreted together. The project is particularly relevant to business intelligence because it connects operational transaction data to management questions. Excel provides transparent inspection and validation, SQL provides repeatable aggregation, and Power BI provides an interactive communication layer.

1.2 ANALYTICAL PRINCIPLE

A core principle throughout the study is traceability. Every headline KPI should be explainable through a clear definition, calculation, and source. This reduces the risk of treating a dashboard visual as an unexplained number.

2. PROBLEM DEFINITION

The central problem addressed by this study is the identification and interpretation of products, divisions, and regions that contribute materially to profitability in the Nassau Candy Distributor dataset. A single aggregate sales figure is insufficient for decision-making. Management needs to know where gross profit is concentrated, which products are leading contributors, whether regional profitability differs, and whether the highest-profit products also demonstrate strong margins. The research problem can therefore be stated as follows: How can structured Excel analysis, SQL-based business questioning, and Power BI visualization be combined to identify and communicate product-line profitability and margin performance in the Nassau Candy Distributor dataset? The project does not attempt to estimate a causal treatment effect or build a production profit-forecasting model. Instead, it creates an auditable descriptive analytics pipeline that can serve as a foundation for future forecasting, pricing analysis, inventory optimization, or customer-level profitability work.

2.1 RESEARCH QUESTIONS

- 1. What are the overall sales, cost, gross profit, units sold, and profit margin?**
- 2. Which division contributes the highest gross profit?**
- 3. Which region contributes the highest gross profit?**
- 4. Which products are the top five contributors to gross profit?**
- 5. How do the leading products compare in sales and profit margin?**
- 6. How can date, region, and division filters support focused investigation?**
- 7. How can the same business metrics be validated across Excel, SQL, and Power BI?**
- 8. Which findings are most relevant for product, regional, and profitability management?**

3. OBJECTIVES, SCOPE AND SIGNIFICANCE

3.1 OBJECTIVES

☒ Validate the supplied transaction data and establish trusted baseline totals. ☒ Measure sales, cost, gross profit, units sold, and gross-profit margin. ☒ Compare gross profit across divisions and regions. ☒ Identify and rank the top five products by gross profit. ☒ Compare leading-product sales with profit margin to avoid a revenue-only interpretation. ☒ Build repeatable SQL queries for the core business questions. ☒ Create Power BI measures and an interactive one-page dashboard. ☒ Translate analytical findings into practical business recommendations.

3.2 SCOPE

☒ Transaction-level sales analysis using the supplied Nassau Candy Distributor dataset. ☒ Financial measures: Sales, Cost, and Gross Profit. ☒ Operational measure: Units Sold.

☒ Dimensions: Order Date, Region, Division, Product Name, and related transaction fields.

☒ Reporting period: 2 January 2024 to 31 December 2025. ☒ Final communication through a one-page Power BI executive dashboard.

3.3 SIGNIFICANCE

The significance of the project lies in demonstrating a complete analyst workflow. The same business definitions are carried from the raw data through Excel, SQL, and Power BI, creating an auditable chain from data to decision support.

4. LITERATURE REVIEW

Profitability analysis is a central component of management accounting and business analytics because organizations need to understand both revenue generation and the costs associated with generating that revenue. Modern business intelligence extends this principle by allowing financial outcomes to be segmented by product, geography, customer, time, and other operational dimensions. Davenport and Harris describe analytics as a strategic capability through which organizations can use data and analysis to improve decision-making and competitive performance. This perspective supports the present project's emphasis on converting transaction records into management-oriented information rather than treating data preparation as an end in itself. Few's work on dashboard design emphasizes that dashboards should communicate critical information efficiently and clearly. The present project follows this principle through a compact KPI strip, focused profitability visuals, a ranked product table, and interactive slicers rather than a large collection of unrelated charts. Exploratory data analysis provides another methodological foundation. Grouping and comparing observations across categories helps analysts discover patterns before more advanced modeling is attempted. In a profitability setting, this includes examining gross profit by division, region, and product and comparing absolute contribution with normalized measures such as margin. SQL provides a structured and repeatable mechanism for aggregating transaction data. MySQL documentation describes aggregate functions such as SUM and COUNT and their use with GROUP BY for analyzing sets of values. This directly supports the project's SQL-based business questions. Microsoft documentation similarly describes Excel PivotTables as tools for calculating, summarizing, and analyzing data to reveal comparisons, patterns, and trends. Power BI provides the final communication layer by allowing measures and filters to be explored interactively.

4.1 RESEARCH GAP AND POSITIONING OF THIS PROJECT

Many portfolio analytics projects stop at either a spreadsheet summary or a dashboard. When the calculation path is not documented, the final visualization can be difficult to audit. This study addresses that practical gap by connecting three analytical layers: Excel validation, SQL aggregation, and Power BI reporting. A second gap concerns revenue-only interpretation. A product with high sales is not necessarily the most attractive product from a profitability perspective. Gross profit incorporates the cost dimension, while profit margin normalizes profitability relative to sales. Viewing these measures together provides a stronger management perspective. A third gap is communication. SQL tables are precise but may not be accessible to non-technical stakeholders. Power BI closes this communication gap by presenting headline KPIs and segment-level comparisons in an interactive visual environment. The study is therefore positioned as a descriptive, business-oriented profitability analytics framework. Its contribution is the integration of data preparation, structured SQL questioning, independent cross-checking, dashboard communication, and business recommendations.

4.2 CONCEPTUAL FRAMEWORK

Analytical stage	Question	Expected output	Data preparation	Can the source be trusted and understood?	Validated dataset
Exploration	Where are the strongest patterns?	Segment summaries	SQL analysis	Can the questions be reproduced?	Reproducible results
Power BI	Can results be communicated interactively?	Dashboard	Interpretation	What do the patterns mean for management?	Findings
Action	What should management investigate or prioritize?	Recommendations			

5. RESEARCH METHODOLOGY

5.1 RESEARCH DESIGN

This study follows a quantitative, descriptive, and cross-sectional analytics design. The unit of analysis is the transaction record, while higher-level summaries are produced by aggregating transactions by division, region, and product. The methodological sequence is: data inspection → data preparation → exploratory analysis → SQL business questions → dashboard development → cross-validation → interpretation → recommendations.

5.2 ANALYTICAL WORKFLOW

Stage Primary activity Output

1. Source

Inspect transaction file and field structure Raw analytical dataset

2. Excel

Validate data, calculate KPIs, explore segments Baseline results

3. MySQL

Aggregate, group, rank, and reconcile Structured results

4. Power BI

Build measures, visuals, and slicers Interactive dashboard

5. Interpretation

Identify patterns and business implications Findings

6. Recommendation

ions Translate findings into actions Decision framework This multi-layer design reduces the risk of relying on a single calculation environment. Excel provides an accessible validation layer, SQL provides reproducibility, and Power BI provides interactive communication.

5.3 FIGURE 1 — END-TO-END ANALYTICS WORKFLOW

Raw Data → Excel Validation → MySQL Analysis → Power BI Dashboard → Business Findings → Recommendations

5.4 ANALYTICAL LAYERS

Layer	Primary purpose	Output
Excel	Preparation, formulas, exploratory summaries, and cross-checking	Validated KPI and segment tables
MySQL	Structured business questions and repeatable aggregation	Query-based analytical results
Power BI	Measures, visuals, filters, and interactive reporting	Decision-oriented dashboard
Business interpretation	Connect results to management questions	Findings and recommendations

5.5 CORE METRIC FORMULAS

Metric

Definition	Total Sales	SUM(Sales)	Total Cost	SUM(Cost)	Total Gross Profit	SUM(Gross Profit)	Total Units	SUM(Units)
Total Orders	DISTINCTCOUNT(Order ID)							

Profit Margin %

$\text{Total Gross Profit} \div \text{Total Sales} \times 100$

For the complete dataset, Profit Margin = $\$93,442.80 \div \$141,783.63 = 65.91\%$.

5.6 VALIDATION STRATEGY

- ☐ Check row count and field structure in Excel.
- ☐ Verify date interpretation and numeric fields.
- ☐ Recalculate core KPIs using spreadsheet formulas or PivotTables.
- ☐ Reproduce the same totals and groupings through SQL.
- ☐ Create Power BI measures from the validated fields.
- ☐ Compare dashboard headline values against the established baseline.
- ☐ Investigate unusual segment values together with their absolute contribution before making recommendations.

6. DATASET DESCRIPTION AND DATA PREPARATION

The supplied dataset contains 10,194 transaction records and 18 fields. Orders cover the period from 2 January 2024 through 31 December 2025. The data combines transaction identifiers, order dates, customer/order attributes, geography, product dimensions, operational volume, and financial measures. Statistic

Value.

Transaction records: 10,194

Unique orders: 8,549

Columns: 18

Reporting period: 2 January 2024 – 31 December 2025

Sales: \$141,783.63

Cost: \$48,340.83

Gross Profit: \$93,442.80

Units Sold: 38,654

6.1 DATA PREPARATION

The preparation stage focused on understanding the source structure, validating numeric measures, interpreting dates correctly, and preserving categorical dimensions such as Region, Division, and Product Name for subsequent grouping. No unsupported predictive model, external market data, or retroactive assumption is introduced into the empirical results. The research paper documents the analytical work actually performed in the project.

6.2 DATA ANALYSIS UNIT

The transaction record is the base unit of analysis. Financial measures are aggregated to answer business questions at the order, division, region, and product levels.

6.3 DATASET FIELD DICTIONARY

Field group Fields Purpose Transaction Row ID, Order ID, Order Date, Ship Date, Ship Mode Transaction tracking and time analysis Customer Customer ID Customer/order relationship context Geography

Country/Region, City, State/Province, Postal Code, Region

Regional performance analysis

Product

Division, Product ID, Product Name

Product mix and profitability

Financial Sales, Cost, Gross Profit Revenue, cost, and profit measurement Operations Units Volume measurement

6.4 BUSINESS METRIC DEFINITIONS

Metric

Business interpretation Sales Total value of sales transactions in the dataset. Cost Total supplied cost associated with the transactions. Gross Profit Profit contribution represented by the supplied Gross Profit field. Units Sold Total number of units represented in the transaction records. Total Orders Distinct count of Order ID.

Profit Margin %

Gross profit expressed as a percentage of sales.

Division

High-level product category used for profitability comparison.

Region

Geographic reporting dimension used for market comparison.

6.5 DATA QUALITY AND VALIDATION

Check Purpose Result Record count Confirm source size Validated Date fields Avoid date ambiguity Validated Numeric measures Confirm Sales, Cost, GP, and Units are numeric Validated Categorical fields

Preserve Region, Division, and Product categories

Validated Core totals Reconcile Excel, SQL, and Power BI Validated

6.6 VALIDATED BASELINE

Measure Validated result Sales

\$141,783.63

Cost

\$48,340.83

Gross Profit

\$93,442.80

Units Sold 38,654 Unique Orders 8,549 Profit Margin

65.91%

Validation is treated as part of the analytical method rather than a final presentation step. A dashboard metric is useful only when its definition and calculation can be explained. The project does not introduce additional missing-value imputation or outlier treatment beyond the documented analytical workflow. This keeps the research paper faithful to the actual project rather than retroactively claiming techniques that were not performed.

7. EXCEL-BASED EXPLORATORY ANALYSIS

Excel was used as the first analytical layer. The purpose was to inspect the source, establish baseline financial totals, calculate derived metrics, and perform exploratory aggregation before database analysis.

7.1 EXCEL WORKFLOW

☒ Import the transaction data into Excel. ☒ Inspect column names, dates, numeric fields, and categorical dimensions. ☒ Check obvious data-quality issues and inconsistent values. ☒ Calculate Sales, Cost, Gross Profit, Units, and Profit Margin.

☒ Use PivotTable-style summaries to compare Division, Region, and Product.

☒ Record baseline results for later SQL and Power BI reconciliation. 7.2 CORE EXCEL KPIs

Metric Value

Total Sales

\$141,783.63

Total Cost

\$48,340.83

Total Gross Profit

\$93,442.80

Total Units

38,654

Unique Orders

8,549

Profit Margin

65.91%

The Excel baseline acts as the reference point for the remainder of the study. Segment-level results are interpreted relative to the overall financial position.

Division

\$88,824.62

95.06%

Other

\$4,333.45

4.64%

Sugar

\$284.73

0.30%

Division

Other

■ ■ \$4.3K

Sugar

■ \$0.3K

7.4 MANAGEMENT INTERPRETATION

The division result suggests that management attention should be concentrated first on protecting the strongest profit engine and second on understanding the commercial and operational reasons behind the much smaller contributions from Other and Sugar.

Region

\$30,485.94

32.63%

\$26,973.70

28.87%

Interior

\$21,282.49

22.78%

Gulf

\$14,700.67

15.73%

Region

 \$30.5K

Atlantic

 \$27.0K

Interior

 \$21.3K

Gulf

■■■■■■■■■■ \$14.7K

Pacific is the leading region in absolute gross profit, followed by Atlantic, Interior, and Gulf.

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8. MYSQL BUSINESS ANALYSIS

The SQL layer transforms exploratory questions into repeatable analytical statements. Instead of manually rebuilding every summary in a spreadsheet, SQL can aggregate the same source table consistently and provide a transparent analytical record.

8.1 SQL DESIGN

The project uses a structured sales table containing order identifiers, dates, geography, product attributes, sales, units, gross profit, and cost. The principal SQL operations are COUNT DISTINCT, SUM, GROUP BY, ORDER BY, CASE expressions, and ranking through ordered aggregation.

8.2 BUSINESS QUESTIONS

No. Business question 1 Overall orders, sales, cost, gross profit, units, and margin. 2 Gross profit by division. 3 Gross profit by region. 4 Top five products by gross profit. 5 Sales and margin of leading products. 6 Performance under selected date, region, and division filters. 7 Reconciliation of SQL totals with Excel. 8 Validation of Power BI measures against SQL results. SQL adds repeatability, scalability, and a clear analytical record. The same business definition can be executed again after a data refresh.

8.3 SQL CORE QUERY EXCERPTS

Overall KPI SELECT COUNT(DISTINCT order_id) AS total_orders, SUM(sales) AS total_sales, SUM(cost) AS total_cost, SUM(gross_profit) AS total_gross_profit, SUM(units) AS total_units FROM candy_sales;

Gross Profit by Division

SELECT division, SUM(gross_profit) AS gross_profit FROM candy_sales GROUP BY division ORDER BY gross_profit DESC;

Gross Profit by Region

SELECT region, SUM(gross_profit) AS gross_profit FROM candy_sales GROUP BY region ORDER BY gross_profit DESC; Top Five Products SELECT product_name, SUM(gross_profit) AS gross_profit FROM candy_sales GROUP BY product_name ORDER BY gross_profit DESC

LIMIT 5;

8.4 VALIDATED SQL RESULTS

Metric

SQL result Distinct Orders 8,549 Sales

\$141,783.63

Cost

\$48,340.83

Gross Profit

\$93,442.80

Units 38,654 Profit Margin

65.91%

8.6 PRODUCT CONCENTRATION

The five products together contribute \$88,824.62 in gross profit. This concentration makes product-level monitoring important for inventory, pricing, promotion, and availability decisions.

9. POWER BI DASHBOARD AND BUSINESS INTELLIGENCE LAYER

Power BI serves as the communication layer of the project. The dashboard is designed to allow a stakeholder to understand overall performance first and then move into division, regional, and product detail.

9.1 DASHBOARD PURPOSE

The final report uses six KPI cards and a focused set of analytical visuals, supported by three slicers: Order Date, Region, and Division. The layout provides a direct reading path from headline financial performance to division and regional profitability and finally to product detail.

9.2 DASHBOARD COMPONENTS

Component Purpose Total Orders Shows distinct order volume. Total Sales Shows revenue scale. Total Cost Shows the supplied cost base. Total Gross Profit Shows absolute profit contribution. Total Units Sold Shows operational volume.

Profit Margin %

Shows profitability efficiency.

Gross Profit by Division

Identifies the dominant product category.

Gross Profit by Region

Compares geographic profitability. Top 5 Products chart Ranks leading profit contributors. Top 5 Products table Combines sales and margin detail. Slicers Enable date, region, and division filtering.

NASSAU CANDY SALES DASHBOARD

KPI	Value
Total Orders	10K
Total Sales	141.78K
Total Cost	48.34K
Total Gross Profit	93.44K
Total Units Sold	39K
Profit Margin %	65.91%

Gross Profit by Division

Division	Gross Profit
Chocolate	89K
Other	4K
Sugar	0K

Gross Profit by Region

Region	Gross Profit
Pacific	30K
Atlantic	27K
Interior	21K
Gulf	15K

Top 5 Products by Gross Profit

Product Name	Gross Profit
Wonka Bar - Scrumdiddlyumptious	19.4K
Wonka Bar - Triple Dazzle Caramel	18.6K
Wonka Bar - Milk Chocolate	17.4K
Wonka Bar - Nutty Crunch Surprise	16.8K
Wonka Bar - Fudge Mallows	16.6K

Top 5 Products - Sales & Margin

Product Rank	ProductName	Division	Total Sales	Profit Margin %
1	Wonka Bar - Scrumdiddlyumptious	Chocolate	27,874.80	69.44%
2	Wonka Bar - Triple Dazzle Caramel	Chocolate	28,485.00	65.33%
3	Wonka Bar - Milk Chocolate	Chocolate	26,867.75	64.92%
4	Wonka Bar - Nutty Crunch Surprise	Chocolate	23,574.95	71.35%
5	Wonka Bar - Fudge Mallows	Chocolate	24,890.40	66.67%

Filters

Filter	Selected / Available
Region	All
Division	All
Order Date	1/2/2024 – 12/31/2025

Figure 6. Final Nassau Candy Sales Dashboard supplied for the project.

The supplied dashboard contains the six headline KPIs: Total Orders, Total Sales, Total Cost, Total Gross Profit, Total Units Sold, and Profit Margin %. It also includes Gross Profit by Division, Gross Profit by Region, Top 5 Products by Gross Profit, and a Top 5 Products table showing Sales and Profit Margin. The visible filters are Order Date, Region, and Division.

The dashboard's displayed headline values are approximately 10K orders, \$141.78K sales, \$48.34K cost, \$93.44K gross profit, 39K units, and 65.91% margin. The underlying project calculations provide the more precise values used throughout this paper.

9.4 POWER BI TECHNICAL BUILD

Core DAX measures
Total Sales = SUM(Candy_Sales[Sales])
Total Cost = SUM(Candy_Sales[Cost])
Total Gross Profit = SUM(Candy_Sales[GrossProfit])
Total Units Sold = SUM(Candy_Sales[Units])
Total Orders = DISTINCTCOUNT(Candy_Sales[OrderID])

Profit Margin % = DIVIDE([Total Gross Profit], [Total Sales], 0)

Product ranking measure

Product Rank =

```
RANKX(
    ALL(Candy_Sales[ProductName]), [Total Gross Profit], ,
    DESC,
    DENSE
)
```

9.5 INTERACTIVE LOGIC

☒ The measures are based on the same fields used in Excel and SQL. ☒ Profit Margin is calculated dynamically so that slicer selections update the ratio. ☒ The ranking measure supports the Top 5 product presentation.

☒ Order Date, Region, and Division slicers provide interactive filtering.

☒ The dashboard headline values reconcile with the validated project totals.

9.6 VISUAL HIERARCHY

The dashboard follows a management-oriented hierarchy: headline KPIs first, division and regional comparison second, and product-level detail third. This hierarchy helps users move from 'What is the overall position?' to 'Where is profit coming from?' and finally to 'Which products should receive attention?'

10. RESULTS AND KEY FINDINGS

10.1 OVERALL FINANCIAL PERFORMANCE

Measure Result Sales

\$141,783.63

Cost

\$48,340.83

Gross Profit

\$93,442.80

Units Sold 38,654 Unique Orders 8,549 Profit Margin

65.91%

The dataset generates \$141,783.63 in sales and \$93,442.80 in gross profit after the supplied cost of \$48,340.83. The resulting 65.91% margin establishes the overall benchmark for segment-level interpretation.

10.2 DIVISION RESULT

Chocolate contributes \$88,824.62, representing approximately 95.06% of total gross profit. The result indicates that the overall profitability position is strongly influenced by one division.

10.3 REGIONAL RESULT

Pacific leads regional gross profit at \$30,485.94. Gulf is the lowest of the four reported regions at \$14,700.67. This gap should be investigated rather than interpreted as evidence of a specific causal factor.

10.4 PRODUCT RESULT

Wonka Bar - Scrumdiddlyumptious is the leading product by gross profit at \$19,357.50. The five leading products are all Wonka Bar products, indicating a strong product-family concentration.

10.5 TOP PRODUCT SALES AND MARGIN

Product

Sales Profit Margin Wonka Bar - Scrumdiddlyumptious

\$27,874.80

69.44%

Wonka Bar - Triple Dazzle Caramel

\$28,485.00

65.33%

Wonka Bar - Milk Chocolate

\$26,867.75

64.92%

Wonka Bar - Nutty Crunch Surprise

\$23,574.95

71.35%

Wonka Bar - Fudge Mallows

\$24,890.40

66.67%

The product table demonstrates why profitability should not be interpreted only through sales. The products have different sales totals and different margin percentages. Wonka Bar - Nutty Crunch Surprise, for example, has a

71.35% margin, while Wonka Bar - Milk Chocolate has a 64.92% margin.

10.6 KEY FINDINGS TABLE

Finding Evidence Management meaning Chocolate dominates profit

\$88.82K GP

Protect and monitor the leading division. Pacific leads regions

\$30.49K GP

Study its product mix and operating practices. Wonka Bars dominate top products Top 5 all Wonka Bars Prioritize leading SKUs for monitoring. Margins differ by product

64.92%–71.35% among top

five Evaluate sales and margin together. Overall margin is strong

65.91%

Use margin as a recurring KPI.

11. DISCUSSION AND BUSINESS INTERPRETATION

The results support a profitability-first interpretation of the Nassau Candy dataset. Overall sales are important, but the most useful management insight is the concentration of gross profit across divisions, regions, and a small set of leading products.

11.1 DIVISION INTERPRETATION

Chocolate is the primary profit engine. Management should protect the availability and profitability of the leading product family while investigating whether lower-performing divisions can be improved through product mix, pricing, promotion, or distribution decisions.

11.2 REGIONAL INTERPRETATION

Pacific leads regional gross profit, while Gulf trails the other three regions. The data alone does not identify why this occurs. Further investigation should compare order volume, product mix, pricing, customer composition, and distribution cost across regions.

11.3 PRODUCT INTERPRETATION

The concentration of the top five products in the Wonka Bar line indicates that product-family performance is a major part of the overall profitability story. Concentration creates both an opportunity and a dependency: strong performance can drive the business, but underperformance in the same family could materially affect overall results.

11.4 MARGIN INTERPRETATION

Profit margin adds context to absolute gross profit. A product can generate high profit because of high sales volume, while another can generate a stronger percentage margin. Both dimensions should therefore be reviewed together.

11.5 ANALYTICAL CAUTION

The observed differences are descriptive. They should be used to prioritize investigation and management attention rather than to claim that product, region, or division membership itself causes profitability.

12. BUSINESS RECOMMENDATIONS

Priority Observed signal Recommended action 1 Chocolate dominates gross profit Protect availability and monitor the performance of leading Chocolate/Wonka Bar products. 2 Pacific leads regional GP Study product mix, pricing, customer composition, and distribution practices for possible replication. 3 Top five products are Wonka Bars Prioritize inventory, promotion, and margin monitoring for leading SKUs. 4

65.91% overall margin

Use margin as a recurring management KPI alongside sales, cost, and units. 5 Gulf trails other regions Investigate whether product mix, order volume, pricing, or distribution efficiency explains the gap. 6 Top-product margins differ Evaluate products using both absolute gross profit and margin. 7 Dashboard supports filtering Use date, region, and division filters for focused monthly or quarterly review.

12.1 RECOMMENDED PROFITABILITY MANAGEMENT FRAMEWORK

Measure → Prioritize → Investigate → Act → Re-measure. Management can first measure current profitability, prioritize high-impact products and regions, investigate the reasons behind observed differences, implement commercial or operational actions, and then compare future results against the baseline.

12.2 FUTURE MONITORING

The dashboard should be refreshed periodically so management can compare current performance with prior periods. A future version should include monthly and quarterly trends, year-over-year comparisons, and additional operational drivers when available.

13. LIMITATIONS AND FUTURE WORK

13.1 LIMITATIONS

⌘ The analysis is descriptive and does not forecast future sales or profit. ⌘ No external competitor, pricing, promotion, inventory, or market datasets were included. ⌘ Gross profit is analyzed as supplied by the source; detailed cost-driver decomposition is outside scope. ⌘ The project does not establish causal relationships between dimensions and profitability. ⌘ The dashboard is a reporting and decision-support product rather than a production optimization system. ⌘ Regional and product differences may be influenced by factors not represented in the dataset.

13.2 FUTURE WORK

⌘ Add monthly and quarterly sales, cost, gross-profit, and margin trends. ⌘ Introduce year-over-year and period-over-period comparisons. ⌘ Add customer-level profitability and customer segmentation. ⌘ Integrate inventory and product-availability data. ⌘ Add pricing and promotion data to evaluate commercial drivers. ⌘ Develop automated refresh from a production database. ⌘ Introduce forecasting after a sufficiently rich time-series dataset is available. ⌘ Build scenario analysis for price, cost, and product-mix changes. ⌘ Evaluate customer lifetime value and contribution margin when additional cost information becomes available. Future analytical sophistication should not remove auditability. New models and dashboards should remain traceable to source data, definitions, assumptions, and business rules.

14. CONCLUSION

This research paper presented an end-to-end descriptive profitability analysis of Nassau Candy Distributor transaction data using Excel, MySQL, and Power BI. The study established \$141,783.63 in sales, \$93,442.80 in gross profit, 38,654 units sold, 8,549 unique orders, and a 65.91% gross-profit margin.

The analysis demonstrates that profitability is concentrated rather than evenly distributed. Chocolate is the dominant division, Pacific is the leading region, and the top five products by gross profit are all Wonka Bar products. The top-product margin comparison further demonstrates that sales volume and profitability efficiency should be considered together. The methodological conclusion is that reliable business intelligence depends on more than a dashboard. Excel provides preparation and baseline validation, SQL provides reproducible business questions, and Power BI provides interactive communication. Together, these layers create an auditable path from raw data to business insight. The principal practical conclusion is that management should protect high-performing product lines, investigate regional differences, monitor product concentration, and use gross profit and margin as recurring performance measures. The project provides a strong foundation for future time-series analysis, forecasting, inventory optimization, customer profitability analysis, and automated dashboard deployment. The study intentionally distinguishes observed results from causal claims. The findings are most appropriately used as evidence for where management should investigate and prioritize attention.

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16. APPENDICES

APPENDIX A — SQL BUSINESS QUESTIONS

1. Overall orders, sales, cost, gross profit, units, and profit margin.
2. Gross profit by division.
3. Gross profit by region.
4. Top five products by gross profit.
5. Sales and profit margin of the leading products.
6. Date-filtered performance analysis.
7. Region-filtered profitability analysis.
8. Division-filtered profitability analysis.
9. Reconciliation of SQL results with Excel.
10. Validation of Power BI measures against the analytical baseline.

APPENDIX B — PROJECT DATASET PROFILE

Statistic

Value

Rows

10,194 Unique orders 8,549

Columns

18 Sales

\$141,783.63

Cost

\$48,340.83

Gross Profit

\$93,442.80

Units 38,654 Profit Margin

65.91%

Date range 2 January 2024 – 31 December 2025

APPENDIX C — FIGURE SOURCE AND REPRODUCIBILITY NOTES

The final dashboard shown in Figure 6 is the supplied Nassau Candy Sales Dashboard rather than a reconstruction. The numerical results in this paper are based on the project's validated analytical outputs. The dashboard displays rounded headline values for presentation, while the research paper retains precise values where appropriate. The analytical chain can therefore be summarized as: source transactions → Excel validation and exploration → MySQL aggregation → Power BI measures and visuals → business interpretation → recommendations.

APPENDIX D — FINAL RESEARCH PAPER CHECKLIST

☑ Front page included. ☑ Declaration included. ☑ Acknowledgement included. ☑ Abstract and keywords included. ☑ Executive summary included. ☑ Table of contents included. ☑ List of figures included.

☒ List of tables included. ☒ Research questions and objectives included. ☒ Literature review included with external references. ☒ Research methodology and validation strategy included. ☒ Dataset description and field dictionary included. ☒ Excel analysis included. ☒ MySQL analysis and SQL excerpts included. ☒ Supplied Power BI dashboard inserted. ☒ Results, discussion, and recommendations included. ☒ Limitations and future work included. ☒ One-page references section included. ☒ Appendices document SQL questions, dataset profile, and reproducibility notes.

— Prepared by **Sonu Kumar** —